

Background

5 million

sites of soil have been contaminated with heavy metals

(Henao and Ghneim-Herrera, 2021)

Zinc oxide exposure has been shown to **reduce earthworm mobility**:

- Decreased speed
- Reduced distance traveled
- Lower burrowing depth

(Bonthala, 2025)



Antioxidants can reduce heavy metal damage, but many are not sustainable or accessible.

Moringa oleifera is a low-cost, fast-growing plant rich in antioxidants that may **help restore mobility** in organisms exposed to toxins (Chauhan et al., 2020).

Purpose

Recognizing *Moringa oleifera*'s potential in increasing earthworm mobility is important as worms are a **vital species** for soil health. Understanding its ability to mitigate the harmful effects of zinc may provide a **sustainable and accessible solution** to protect soil organisms.

Variables and Hypothesis

IV

Moringa oleifera
(0.1%, 0.5%,
1% w/w)

DV

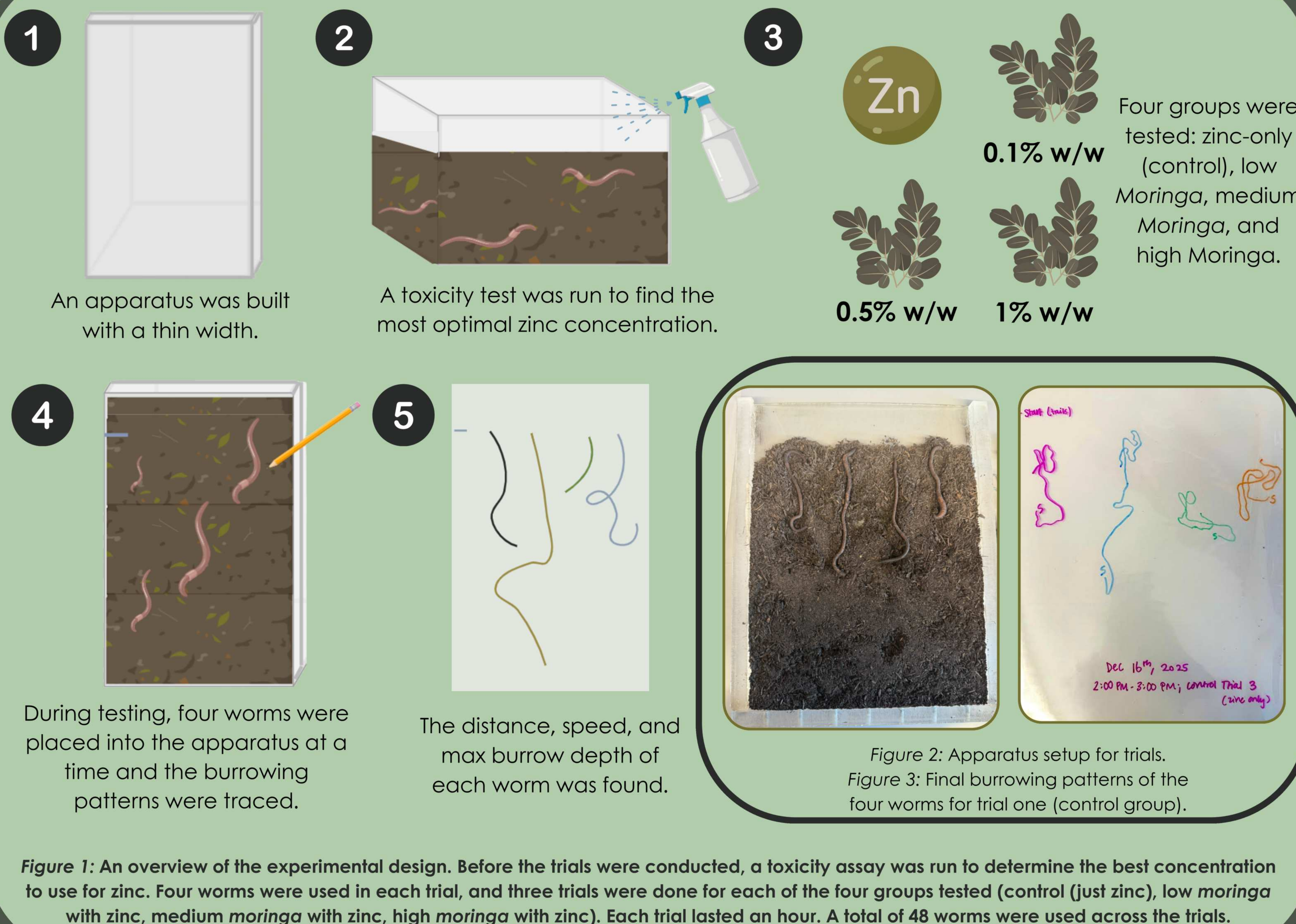
Mobility
(distance,
speed, max
burrow depth)

Control

Mobility of
worms
exposed to
only zinc (no
moringa)

Treating worms with ***Moringa oleifera*** after **zinc** oxide exposure would lead to **increased mobility**, compared to worms exposed to zinc alone.

Testing *Moringa oleifera* as a Potential Treatment in Combatting the Negative Effect of Zinc in Soil on the Mobility of *Lumbricus terrestris*



Results

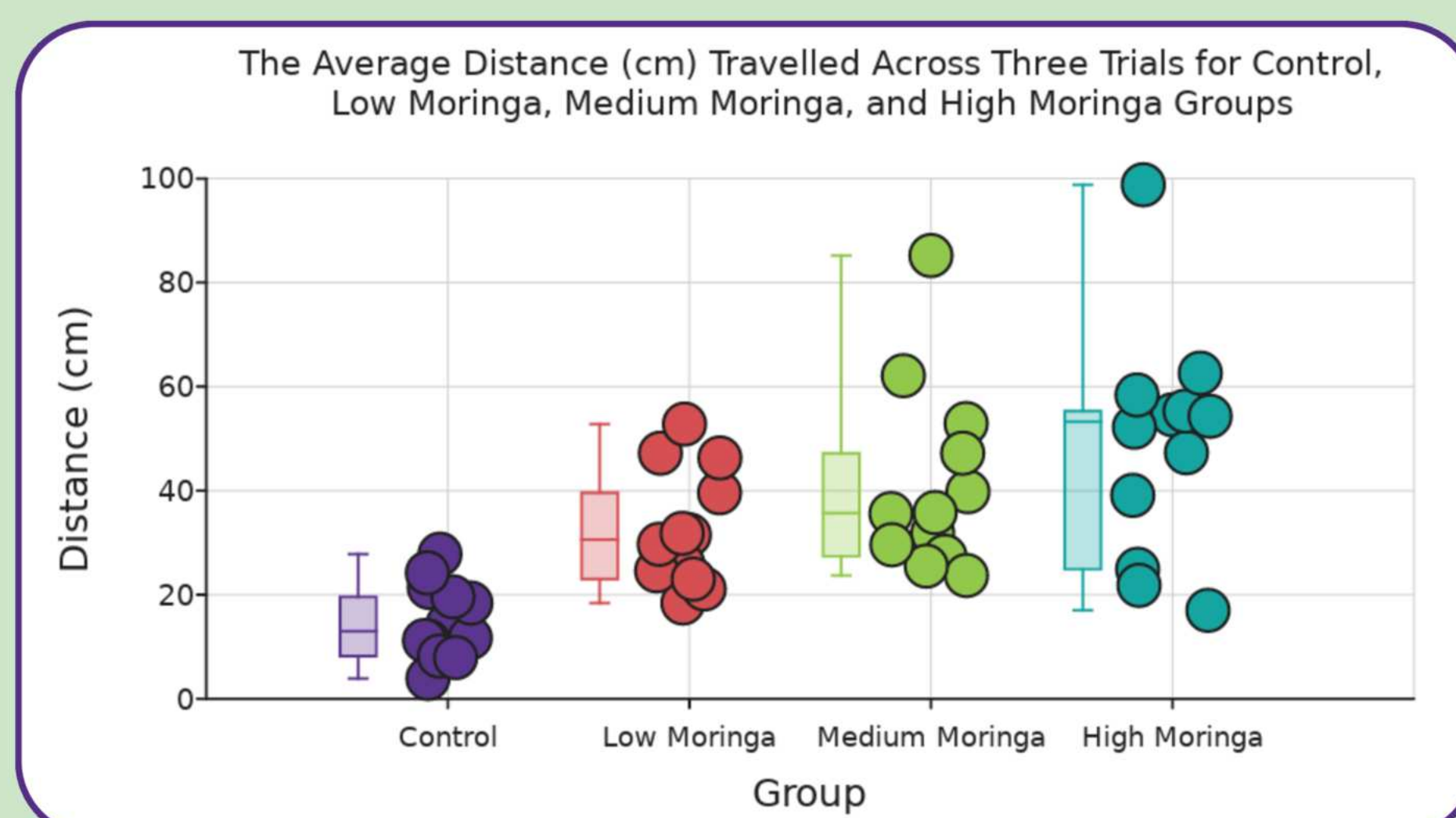


Figure 3: A graph displaying the average distance (cm) travelled across the three trials for the control, low moringa, medium moringa, and high moringa groups is displayed. Each point represents one worm's distance.

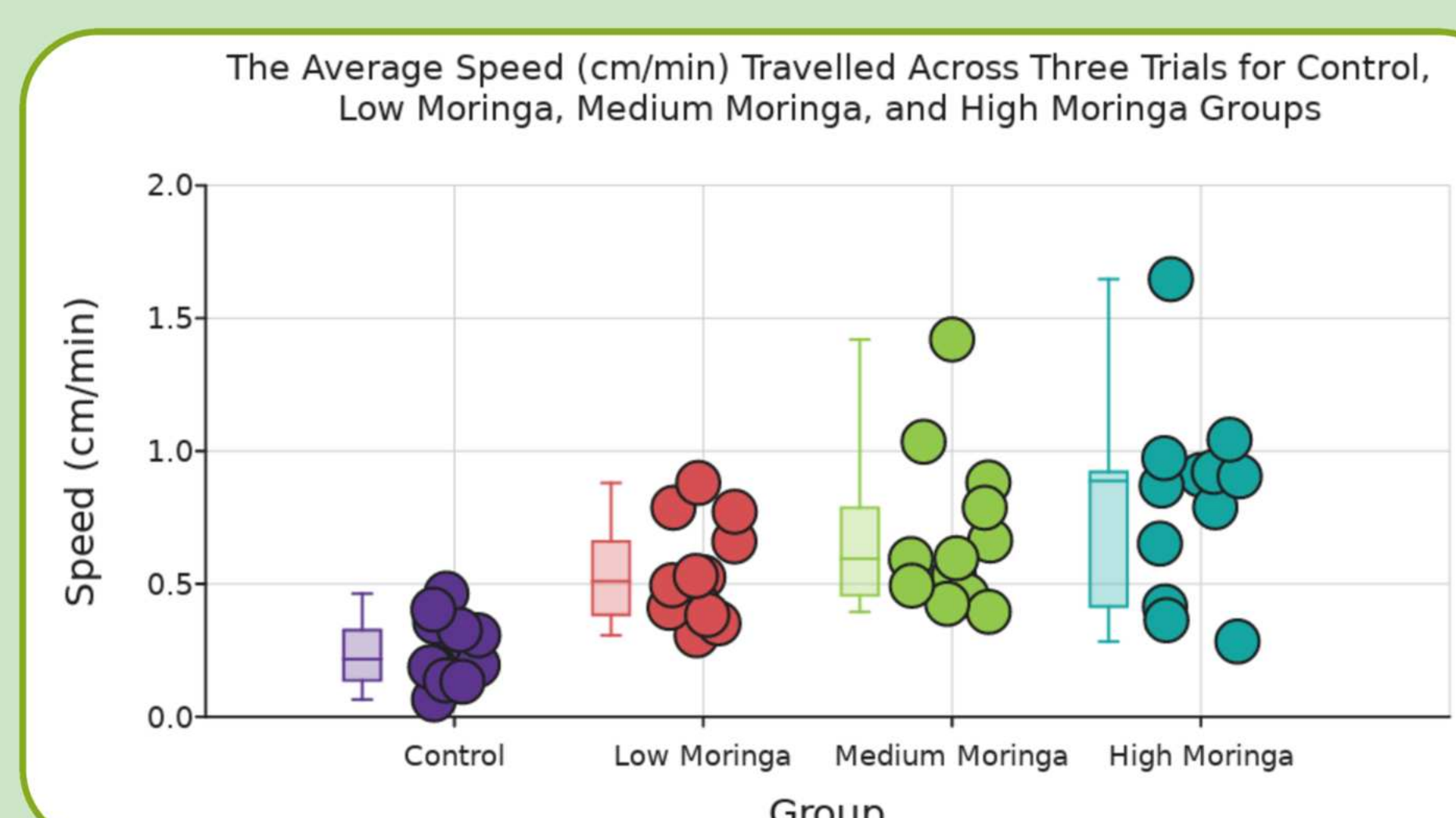


Figure 4: A graph displaying the average speed (cm/min) across the three trials for the control, low moringa, medium moringa, and high moringa groups is displayed on the right. Each point represents one worm's speed.

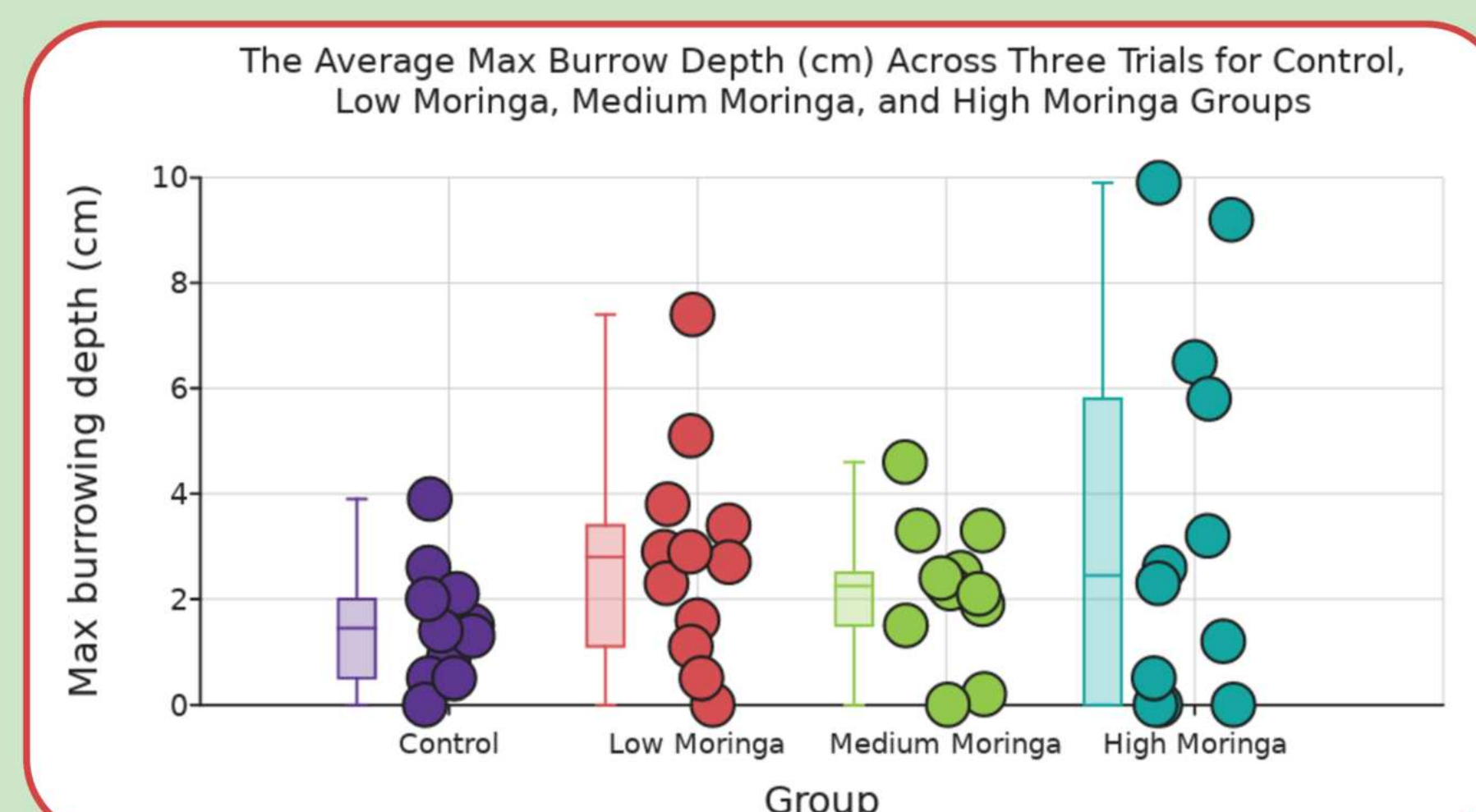


Figure 5: A graph displaying the average max burrow depth (cm) across the three trials for the control, low moringa, medium moringa, and high moringa groups is displayed on the left. Each point represents one worm's speed. An ANOVA test was run and the p-value was 0.20 (little to no indication that the groups are different).

Figures and images were created by the researcher (either documented or through BioRender). Some images were retrieved from Canva. All graphs were created by the researcher using DataClassroom. The statistics tests were run in DataClassroom.

Discussion

The data collected **supported the hypothesis** that exposure to moringa increases earthworm's mobility *to an extent*.

For distance and speed, there was **strong evidence that the groups differed**.

- Found significant differences between the control and low, medium, and high moringa groups.
- Were no significant differences between the moringa treatment groups.

For maximum burrow depth, the post-hoc test showed **little to no evidence that the groups differed**.

Limitations:

- A limited sample size was used, with only twelve data points collected per group.
- Only three mobility factors (distance, speed, and maximum burrow depth) were tested in the experiment.

Implications and Future Work

- **Moringa exposure** appears to **increase certain factors of earthworm mobility**, which may improve soil aeration and nutrient cycling.
- **Future studies** could increase sample size, test additional mobility factors (ex. surface avoidance, coiling), and examine long-term exposure effects.

References

- Bonhala, S. (2025). *Measuring the Effect of Zinc and Lead in Soil on the Mobility of Lumbricus terrestris* [Unpublished manuscript]. Academies of Science, Academies of Loudoun.
- Chauhan, A. P., Chaubey, M. G., Patel, S. N., Madamwar, D., & Singh, N. K. (2020). Extension of life span and stress tolerance modulated by DAF-16 in *Caenorhabditis elegans* under the treatment of *Moringa oleifera* extract. *3 Biotech*, 10(12), 504. <https://doi.org/10.1007/s13205-020-02485-x>.
- González Henao, S., & Ghneim-Herrera, T. (2021). Heavy metals in soils and the remediation potential of bacteria associated with the plant microbiome. *Frontiers in Environmental Science*, 9. <https://doi.org/10.3389/fenvs.2021.604216>.